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Reworkable zero-force insertion electrical optical package socket and method

Abstract

An electronic device comprises an electro-optical circuit package including at least photonic integrated circuit (PIC) having at least one light source and a package substrate; a printed circuit (PCB) including at least one optical connector to receive light from the at least one light source; and multiple liquid metal electrical contacts disposed between the package substrate and the PCB.

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Background/Summary

TECHNICAL FIELD

(1) Embodiments pertain to packaging of photonic integrated circuits (PICs). Some embodiments relate to techniques to couple an optical signal from a PIC to a waveguide or optical fiber.

BACKGROUND

(2) A photonic integrated circuit (PIC) can generate an optical signal. Optically coupling the optical signal to a waveguide allows the optical signal to be used in an optical interface that could be used as a high-speed interface between electronic devices. A conventional approach to optically couple a PIC and a fiber or waveguide involves careful alignment and gluing of the optical fibers to the PIC. This approach can be very challenging for high volume manufacturing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a side view illustration of an example of an assembly including an electro-optical circuit package in accordance with some embodiments;

(2) FIG. **2** is a conceptual block diagram of an example of an optical alignment system in accordance with some embodiments;

(3) FIG. **3** is a side view illustration of another example of an assembly including an electro-optical circuit package in accordance with some embodiments;

(4) FIGS. **4**A-**4**B are illustrations of an example of a method to automatically socket an electro-optical circuit package in accordance with some embodiments;

(5) FIGS. **5**A-**5**B are illustrations of an example of a method to de-socket the electro-optical circuit package in accordance with some embodiments;

(6) FIG. **6** is a side view illustration of another example of an assembly that includes an electro-optical circuit package in accordance with some embodiments;
(7) FIG. **7** is a block diagram of an example of an optical alignment system for mounting an electro-optical circuit package to a PCB in accordance with some embodiments;
(8) FIG. **8** is a side view illustration of another example of an assembly that includes an electro-optical circuit package in accordance with some embodiments;
(9) FIG. **9** is an illustration of an example of another application for the optical alignment system of FIG. **7** in accordance with some embodiments;
(10) FIG. **10** is a flow diagram of an overview of a method of forming an electronic device in accordance with some embodiments;
(11) FIG. **11** illustrates a system level diagram in accordance with some embodiments.

DETAILED DESCRIPTION

(12) The following description and the drawings sufficiently illustrate specific embodiments to enable those skilled in the art to practice them. Other embodiments may incorporate structural, logical, electrical, process, and other changes. Portions and features of some embodiments may be included in, or substituted for, those of other embodiments. Embodiments set forth in the claims encompass all available equivalents of those claims.

(13) Conventional techniques to incorporate optical signaling into electronic packages make high volume manufacturing challenging. These techniques can involve direct coupling of an optical fiber to a photonic integrated circuit (PIC). For example, optical fibers can be glued into V-grooves etched into the edge of the PIC. These direct coupling techniques can involve tight positional tolerances making alignment by the assembly tools challenging. This can result in low throughput and makes the conventional techniques unsuitable for high volume manufacturing.

(14) FIG. **1** is a side view illustration of an example of an assembly **100** involving an optical coupling of an optical waveguide and a photonic integrated circuit (PIC) that is scalable to high volume manufacturing. The assembly **100** includes an electro-optical circuit package **101** mounted on a package substrate **104**. The electro-optical circuit package **101** includes one or more integrated circuits (not shown). The integrated circuits include the PIC and can include one or more processors (e.g., a central processing unit (CPU), graphics processing unit (GPU), etc.) and memory (e.g., high bandwidth memory (HBM), etc.). The PIC can include a light source (e.g., a laser diode). The electro-optical circuit package **101** also includes an integrated heat spreader (IHS) **106**. A heat sink **108** can be attached to the IHS **106** using clamp **110**.

(15) The electro-optical circuit package **101** is placed in a package socket **112** that is mounted on a printed circuit board (PCB) **114**. The package socket **112** may be a pin socket that includes pins **116** to contact the package substrate **104**, or the PCB includes the pins **116** and the pins **116** pass through opening of the package socket **112**. The package socket **112** or PCB **114** also includes longer pins **118** that include one or more of power, ground, and control pins to contact the package substrate **104**. The electro-optical circuit package **101** includes liquid metal electrical contacts **120** to contact the pins **116**, **118**. The liquid metal electrical contacts **120** may be included on the circuit package substrate **104** or the liquid metal electrical contacts **120** may be included in an interposer **122** contacting the circuit package substrate **104**. In some examples, the interposer **122** may be placed on the package socket **112** separately before the circuit package substrate **104** is placed on the package substrate. The circuit package substrate **104** may include pins or pads (not shown) to contact the interposer **122**.

(16) For complex integrated circuit designs, the number of connections needed between a circuit package and the package socket can become large. This can lead to a high insertion force needed in conventional attachment techniques. The liquid metal electrical contacts **120** allow for a zero-insertion force package. The liquid metal of the liquid metal electrical contacts **120** can be any metal that has a melting point at or near room temperature. Some examples include cesium, gallium, and rubidium. In some examples, the liquid metal is an alloy of gallium and indium, The

liquid metal may also be a eutectic that is an alloy having a melting point at or near room temperature. In some examples, a sealing film, or cap seal, may be used to hold the liquid metal in place until attachment is made. An interposer **122** may include a cap seal on both the top and bottom sides of the interposer. In some examples, the liquid metal has a zero flow rate, and a cap seal is not needed.

(17) The assembly **100** also includes an optical connector **124** connected to a waveguide **126**. The waveguide **126** is shown embedded in the PCB **114** but the waveguide **126** may be external to the PCB **114**. An optical signal (shown as λ) from the light source of the IC is used to align the circuit package and the package socket **112**. A positioning mechanism (e.g., a six-axis positioning mechanism for x, y, z, pitch, yaw, and roll) can adjust the position of the electro-optical circuit package **101** according to the optical signal. For example, the positioning mechanism may move the electro-optical circuit package **101** to find the position with the maximum light intensity from the light source and attach the circuit package to the package socket **112** when the optimized position is found.

(18) To provide power to the PIC light source during alignment, the power and ground pins **118** are longer than the other pins **116**. The electro-optical circuit package **101** contacts the power and ground pins **118** first to power the light source before the other pins **116** contact the circuit package. The optimum alignment of the powered light source and the optical connector is then found. The positioning mechanism then lowers the circuit package to complete the connection between pins **116** and the liquid metal electrical contacts **120**. An adhesive may be used to complete the attachment. The positioning mechanism may hold the circuit package until adhesive cures.

(19) As explained previously herein, the careful alignment and gluing of the optical fibers to the PIC in a conventional attachment technique can be challenging. In the technique of the example of FIG. **1**, powering the light source provides feedback for automatic alignment of the light source and optical connector **124**. The liquid metal contact of the power pin provides a flexible power connection for the six-axis movement of the positioning mechanism while the optimum alignment is found. The alignment algorithm may continue to run during the cure process.

(20) FIG. **2** is a block diagram of an example of an optical alignment system **200** for mounting an electro-optical circuit package **101** to a PCB **114**. The functional parts of the system **200** are shown in block form without attempting to show the actual relative sizes of the functional parts. The system **200** includes a six-axis positioning mechanism **240** to align the electro-optical circuit package **101** and the PCB **114**. The system **200** includes a positioning circuit **244** that includes a light sensor **248**, logic circuitry **250**, and power and control (PC) interface **252**.

(21) Active feedback for the positioning mechanism **240** is provided by the optical signal from the light source **242** of the PIC. The power and control interface **252** provides power to the light source **242** of the PIC through the pins **118** of the PCB **114**. The optical connector **124** receives the optical signal from the light source **242** which is detected on waveguide **126** by the light sensor **248** (e.g., a photo diode). The logic circuitry **250** provides one or more positioning control signals to drivers **246** of the six-axis positioning mechanism **240** in response to the optical signal. In some examples, one or more of the light sensor **248**, logic circuitry **250**, and power and control interface **252** can be included on the PCB board **114**.

(22) The logic circuitry **250** may perform an algorithm to move the position of the electro-optical circuit package **101** to maximize the light intensity of the optical signal. The control signals from the positioning circuit **244** may cause the drivers **246** to move the positioning mechanism **240** in movement search windows of dimensions of 100 microns (100 μm) or less to find the position of the electro-optical circuit package **101** with optimized alignment. A user interface **254** (UI) may be provided to make changes to the algorithm performed by the logic circuitry **250**. In an alternative, the positioning mechanism **240** may be attached to the PCB **114** and the logic circuitry **250** moves the position of the PCB **114** relative to the electro-optical circuit package **101** to maximize the light intensity of the optical signal.

(23) The light source **242** of the PIC may include any device that generates an optical signal (e.g., a laser emitting diode, a light emitting diode (LED), etc.). In the example of FIGS. **1** and **2**, the optical signal is emitted vertically from the circuit package (e.g., a vertical-emitting diode) and the optical connector **124** is arranged vertically from the circuit package for a bottom optical signal coupling.

(24) FIG. **3** is a side view of another assembly **300** in which the light source of the PIC is a side-emitting diode. The PCB **314** includes the optical connector **328** mounted to the side of the circuit package for an edge optical signal coupling. The waveguide **330** connected to the optical connector **328** is external to the PCB **314** in the example of FIG. **3**. The process of mounting the electro-optical circuit package **101** to the PCB **314** in FIG. **3** is similar to the process described regarding FIG. **1**.

(25) FIGS. **4**A-**4**B are illustrations of an example of a method to automatically socket the electro-optical circuit package **101** of FIG. **1** into the package socket **112**. A removable frame or set of one or more removable tabs **432** is added to the electro-optical circuit package **101**. The removable tabs **432** or removable frame contact one or both of the package substrate **104** and the IHS **106**. The removable tabs **432** may be a removable frame that extends around the circumference of the electro-optical circuit package **101** and IHS **106**, or the removable tabs **432** may extend around only a portion of the circumference. One or more temporary fasteners **434** (e.g., a screw, pin, etc.) may be used to attach the one or more tabs to the package substrate **104** or the IHS **106**. One or more removable landing pads **436** for the one or more removable tabs **432** is added to the package socket **112**. One or more temporary fasteners **434** may be used to attach the landing pads **436** to the package socket **112**.

(26) To socket the electro-optical circuit package **101**, an adhesive **438** can be applied to one or both of the removable tabs **432** and the removable landing pads **436**. The electro-optical circuit package **101** and removable tabs **432** are lowered toward the package socket **112** and removable landing pads **436** to an alignment separation so that the liquid metal electrical contact **120** contact at least one of a power or ground pin **118** to provide power to the light source of the PIC. Alignment is then performed to optimize the optical signal from the light source received by the optical connector **124**.

(27) FIG. **4**B shows the electro-optical circuit package **101** and removable tabs **432** lowered onto the package socket **112** and landing pads **436**. The liquid metal electrical contacts **120** contact the pins **116** of the PCB **114** or package socket **112**, and the light source has optimized alignment to the optical connector **124**. The adhesive **438** can be cured (e.g., thermal curing, or ultraviolet (UV) light curing, etc.). The optical alignment system may hold the electro-optical circuit package **101** in place while the adhesive cures to hardening.

(28) FIGS. **5**A-**5**B are illustrations of an example of a method to de-socket the electro-optical circuit package **101** of FIG. **1** from the package socket **112**. In FIG. **5**A the temporary fasteners **434** are removed. In FIG. **5**B, the removable tabs **432** and removable landing pads **436** are removed. Because the electrical contacts are liquid metal electrical contacts and not hardened solder, the electro-optical circuit package **101** can be lifted from the package socket **112**.

(29) The removable tabs **432** and removable landing pads **436** may be removed together. In some examples, the hardened adhesive **438** may be dissolved or broken during the de-socketing and the removable tabs **432** and removable landing pads **436** removed separately. The removable tabs **432** and removable landing pads **436** may be discarded or cleaned and reused.

(30) Other arrangements of the elements of the electro-optical circuit package **101** and the package socket **112** are possible. For example, the liquid metal electrical contacts **120** may be included on the package socket **112**. The package substrate **104** of the electro-optical circuit package **101** may include the pins **118** and **116** for contacting the liquid metal electrical contacts **120**. In another arrangement, the package socket **112** of the PCB **114** is adjusted using the optical alignment system to align the optical connector **124** of the package socket **112** and the light source of the PIC.

(31) FIG. **6** is a side view illustration of another example of an assembly **600** that includes an electro-optical circuit package **601**. The electro-optical circuit package **601** includes one or more PICs and is similar to the electro-optical circuit package **101** of FIG. **1**. The electro-optical circuit package **601** includes package substrate **104**, an IHS **106**, heat sink **108**, and clamp **110** to hold heat sink to the IHS **106**. The assembly **600** includes a packet socket **612** having multiple zones that are movable relative to each other. The example in FIG. **6** shows two socket zones (Zone 1 and Zone 2 labeled as **613**A, **613**B, respectively) but a package socket may have more than two socket zones (e.g., four socket zones). The multi-zoned package socket **612** is a pin socket that includes pins **116** and **118**. The assembly **600** includes two sets of liquid metal electrical contacts. A first set of liquid metal electrical contacts **120** is shown attached to the electro-optical circuit package **601** and a second set of liquid metal electrical contacts **121** included with the PCB **614**. One or both sets of liquid metal electrical contacts may include a sealing film **660**. As noted in regard to the example of FIG. **1**, the first set of liquid metal electrical contacts **120** may be included in a separate interposer.
(32) The electro-optical circuit package **602** includes two light sources, one for each socket zone. The light source can be included in one or two PICs of the electro-optical circuit package **602**. The electro-optical circuit package **602** may include more than one light source for one socket zone. Each socket zone includes pins **118** which include one or both of a power pin and a ground pin longer than pins **116** to contact the first set of liquid metal electrical contacts **120** without the other pins **116** contacting the first set of liquid metal electrical contacts **120**. The pins **118** of the socket zone provides power for the light source corresponding to the socket zone. This allows the light sources to be powered with an adjustment separation between the electro-optical circuit package **601** and the multi-zoned package socket **612**.
(33) Each socket zone **613**A, **613**B includes an optical connector **124**A, **124**B. Each optical connector is connected to a waveguide **130**A, **130**B external to the PCB **614**. The optical connector of the socket zone receives an optical signal (shown as λ.sub.1, λ.sub.2) from the light source corresponding to the socket zone. Each socket zone is positioned to optimize the optical signal for its socket zone. In some examples, each socket zone **613**A, **613**B can include more than one light source, optical connector pair. In this case, each socket zone may be positioned for an overall optimization of the intensity of the multiple optical signals corresponding to the socket zone.
(34) FIG. **7** is a block diagram of an example of an optical alignment system **700** for mounting an electro-optical circuit package (e.g., electro-optical circuit package **601** in FIG. **6**) to a PCB **614** having a multi-zoned package socket (e.g., multi-zoned package socket **612** in FIG. **6**). In the example of FIG. **7**, the optical alignment system **700** includes two six-axis positioning mechanisms **740**A, **740**B, to independently position the two socket zones **613**A, **613**B of the multi-zoned package socket **612**. The optical alignment system **700** may include more than two positioning mechanisms to align more than two socket zones (e.g., four positioning mechanisms to independently align four socket zones). The positioning mechanisms **740**A, **740**B independently align the socket zones **613**A, **613**B to the light sources **742**A, **742**B of the electro-optical circuit package **601**.
(35) The optical alignment system **700** includes a positioning circuit **744** that includes logic circuitry **650**, and a light sensor and a power and control interface for each socket zone. The power and control interfaces **252**A, **252**B of each socket zone **613**A, **613**B provide power to the respective light source **742**A, **742**B through the pins **118** of the socket zone. The system **700** also includes separate drivers **746**A, **746**B for each of the two positioning mechanisms **740**A, **740**B.
(36) Active feedback for the positioning mechanism is provided for each socket zone while the socket zone is at the adjustment separation from the electro-optical circuit package **601**. For Zone 1, an optical signal from light source **742**A is provided to optical connector **124**A which is detected on waveguide **130**A by light sensor **748**A. The logic circuitry **750** provides one or more positioning control signals to drivers **746**A of the six-axis positioning mechanism **740**A in response to the optical signal. The logic circuitry **750** adjusts the position of the socket zone **613**A to optimize the

optical signal from light source **742**A. Similarly for Zone 2, an optical signal from light source **742**B is provided to optical connector **124**B which is detected on waveguide **130**B by light sensor **248**B. The logic circuitry **750** provides one or more positioning control signals to drivers **746**B of the six-axis positioning mechanism **740**B in response to the optical signal. The logic circuitry **750** adjusts the position of the socket zone **613**B to optimize the optical signal from light source **742**B.
(37) The control signals from the positioning circuit **744** may cause the drivers **746**A, **746**B to move the positioning mechanisms **740**A, **740**B, in movement search windows of dimensions of 100 μm or less to find the position of each socket zone **613**A, **613**B with optimized alignment. When the optimum alignment is found, the electro-optical circuit package **601** is attached to the multi-zone socket **612** and the PCB **614** with the other pins **116** contacting the liquid metal electrical contacts **120**.
(38) FIG. **8** is a side view illustration of another example of an electronic assembly **800** that includes an electro-optical circuit package. The example of FIG. **8** shows removable features for attaching the electro-optical circuit package **601**, multi-zoned package socket **612**, and PCB **614** of FIG. **6**. Removable frame **832** contacts one or both of the package substrate **604** and the IHS **106**, and may be a frame around the circumference of the electro-optical circuit package **601** and IHS **106**, or around only a portion of the circumference. Temporary fasteners **834** may be used to attach the frame to the package substrate **604** or the IHS **106**. The removable frame **832** secures the attachment of both the electro-optical circuit package **601** and the package socket **612** to the PCB **614**. A first removable landing pad **836** for the removable frame **832** is included on the package socket **612**, and a second removable landing pad **856** is included on the PCB **614**. The landing pads **836**, **856** may extend around the circumference of the electro-optical circuit package **601** and IHS **106**, or around only a portion of the circumference. The landing pads **836**, **856** may include multiple sections. One or more temporary fasteners **834** may be used to attach landing pad **836** to the package socket **612**, and to attach landing pad **856** to the PCB **614**. Adhesive **838** can be applied to one or more of the removable frame **832** and landing pads **836**, **856**. The electro-optical circuit package **601** can be de-socketed using any of the methods described regarding FIGS. **5**A-**5**B.
(39) FIG. **9** is an illustration of another application for the optical alignment system **700** of FIG. **7**. An array **960** of electro-optical circuit packages are linked to a common thermal solution **968**. Thermal solution **968** may include a heat sink in contact with both electro-optical circuit packages. The example shows two electro-optical circuit packages **901**A, **901**B, but the array **960** can include more than two electro-optical circuit packages. The two electro-optical circuit packages **901**A, **901**B can be similar to the electro-optical circuit package **101** of FIG. **1**, and may include integrated circuits mounted on a package substrate, an IHS and liquid metal electrical contacts.
(40) The two electro-optical circuit packages **901**A, **901**B are mounted to a PCB **914** using a package socket having two zones **613**A, **613**B as in the example of FIG. **6**. Pins **118** provide power to light sources **942**A, **942**B of the electro-optical circuit packages **901**A, **901**B, respectively. The positioning mechanisms **740**A, **740**B, independently position the two socket zones **613**A, **613**B by aligning optical connectors **124**A, **124**B of the socket zones **613**A, **613**B to the light sources **942**A, **942**B. When the optimum alignment is found by the positioning circuit **744**, the electro-optical circuit packages **901**A, **901**B are attached to the multi-zone package socket and the PCB **914** to provide electrical contact to the liquid metal electrical contacts of the electro-optical circuit package **901**A. **901**B. A removable frame may secure the array **960** of electro-optical circuit packages to the PCB **914**. The frame may extend around the circumference of the array **960** of electro-optical circuit packages or around only a portion of the circumference.
(41) For completeness, FIG. **10** is a flow diagram of an overview of a method **1000** of forming an electronic device, such as the electronic device in any of the examples of FIGS. **1-6** and **8**. At block **1005**, a PIC is arranged on a package substrate of an electro-optical circuit package. The PIC includes at least one light source. At block **1010**, multiple liquid metal electrical contacts are

disposed on the package substrate. In certain examples, the liquid metal electrical contacts are included in an interposer disposed on the package substrate. The electro-optical circuit package is aligned and attached to a PCB. The alignment and attachment may be formed by any of the example devices of FIGS. **7** and **9**.

(42) At block **1015**, light from the at least one light source of the PIC is received using an optical connector included on a PCB. At block **1020**, the position of the package substrate is aligned to a location on the PCB by aligning the light source of the PIC and the optical connector of the PCB. At block **1025**, the package substrate is attached to the PCB.

(43) The electro-optical circuit packages may be included in an optical interface between two or more higher level electronic devices. Using zero-force liquid metal electrical contacts for the devices allows for automatic positioning of the optical elements that is scalable and still provides optimization of the optical coupling. The result is circuit package that is removable for maintenance. An example of a higher level electronic device using assemblies with electro-optical circuit packages as described in the present disclosure is included to show an example of a higher level device application.

(44) FIG. **11** illustrates a system level diagram, according to one embodiment of the invention. For instance, FIG. **11** depicts an example of an electronic device (e.g., system) that can include one or more electro-optical circuit packages as described in the present disclosure. In one embodiment, system **1100** includes, but is not limited to, a desktop computer, a laptop computer, a netbook, a tablet, a notebook computer, a personal digital assistant (PDA), a server, a workstation, a cellular telephone, a mobile computing device, a smart phone, an Internet appliance or any other type of computing device. In some embodiments, system **1100** is a system on a chip (SOC) system. In one example, two or more systems as shown in FIG. **11** may be coupled together with an optical interface implemented using one or more electro-optical circuit packages as described in the present disclosure.

(45) In one embodiment, processor **1110** has one or more processing cores **1112** and **1112**N, where N is a positive integer and **1112**N represents the Nth processor core inside processor **1110**. In one embodiment, system **1100** includes multiple processors including **1110** and **1105**, where processor **1105** has logic similar or identical to the logic of processor **1110**. In some embodiments, processing core **1112** includes, but is not limited to, pre-fetch logic to fetch instructions, decode logic to decode the instructions, execution logic to execute instructions and the like. In some embodiments, processor **1110** has a cache memory **1116** to cache instructions and/or data for system **1100**. Cache memory **1116** may be organized into a hierarchal structure including one or more levels of cache memory.

(46) In some embodiments, processor **1110** includes a memory controller **1114**, which is operable to perform functions that enable the processor **1110** to access and communicate with memory **1130** that includes a volatile memory **1132** and/or a non-volatile memory **1134**. In some embodiments, processor **1110** is coupled with memory **1130** and chipset **1120**. Processor **1110** may also be coupled to a wireless antenna **1178** to communicate with any device configured to transmit and/or receive wireless signals. In one embodiment, the wireless antenna interface **1178** operates in accordance with, but is not limited to, the IEEE 802.11 standard and its related family, Home Plug AV (HPAV), Ultra-Wide Band (UWB), Bluetooth, WiMax, or any form of wireless communication protocol.

(47) In some embodiments, volatile memory **1132** includes, but is not limited to, Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS Dynamic Random Access Memory (RDRAM), and/or any other type of random access memory device. Non-volatile memory **1134** includes, but is not limited to, flash memory, phase change memory (PCM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), or any other type of non-volatile memory device.

(48) Memory **1130** stores information and instructions to be executed by processor **1110**. In one

embodiment, memory **1130** may also store temporary variables or other intermediate information while processor **1110** is executing instructions. In the illustrated embodiment, chipset **1120** connects with processor **1110** via Point-to-Point (PtP or P-P) interfaces **1117** and **1122**. The interfaces **1117** and **1122** may include one or more optical interfaces. Chipset **1120** enables processor **1110** to connect to other elements in system **1100**. In some embodiments of the invention, interfaces **1117** and **1122** operate in accordance with a PtP communication protocol such as the Intel® QuickPath Interconnect (QPI) or the like. In other embodiments, a different interconnect may be used.

(49) In some embodiments, chipset **1120** is operable to communicate with processor **1110**, **1105**N, display device **1140**, and other devices **1172**, **1176**, **1174**, **1160**, **1162**, **1164**, **1166**, **1177**, etc. Buses **1150** and **1155** may be interconnected together via a bus bridge **1172**. Chipset **1120** connects to one or more buses **1150** and **1155** that interconnect various elements **1174**, **1160**, **1162**, **1164**, and **1166**. Chipset **1120** may also be coupled to a wireless antenna **1178** to communicate with any device configured to transmit and/or receive wireless signals. Chipset **1120** connects to display device **1140** via interface (I/F) **1126**. Display **1140** may be, for example, a liquid crystal display (LCD), a plasma display, cathode ray tube (CRT) display, or any other form of visual display device. In some embodiments of the invention, processor **1110** and chipset **1120** are merged into a single SOC. In one embodiment, chipset **1120** couples with (e.g., via interface **1124**) a non-volatile memory **1160**, a mass storage medium **1162**, a keyboard/mouse **1164**, and a network interface **1166** via I/F **1124** and/or I/F **1126**, I/O devices **1174**, smart TV **1176**, consumer electronics **1177** (e.g., PDA, Smart Phone, Tablet, etc.). One or more of interfaces **1124** and **1126** may be an optical interface implemented using one or more of the electro-optical circuit packages described herein.

(50) In one embodiment, mass storage medium **1162** includes, but is not limited to, a solid state drive, a hard disk drive, a universal serial bus flash memory drive, or any other form of computer data storage medium. In one embodiment, network interface **1166** is implemented by any type of well-known network interface standard including, but not limited to, an Ethernet interface, a universal serial bus (USB) interface, a Peripheral Component Interconnect (PCI) Express interface, a wireless interface and/or any other suitable type of interface. In one embodiment, the wireless interface operates in accordance with, but is not limited to, the IEEE 802.11 standard and its related family, Home Plug AV (HPAV), Ultra-Wide Band (UWB), Bluetooth, WiMax, or any form of wireless communication protocol.

(51) While the modules shown in FIG. **11** are depicted as separate blocks within the system **1100**, the functions performed by some of these blocks may be integrated within a single semiconductor circuit or may be implemented using two or more separate integrated circuits. For example, although cache memory **1116** is depicted as a separate block within processor **1110**, cache memory **1116** (or selected aspects of **1116**) can be incorporated into processor core **1112**.

(52) The devices, systems, and methods described can provide improved routing of interconnection between ICs for a multichip package in addition to providing improved transistor density in the IC die. Examples described herein include two or three IC dies for simplicity, but one skilled in the art would recognize upon reading this description that the examples can include more than three IC dice.

ADDITIONAL DESCRIPTION

(53) Example 1 includes subject matter (such as an electronic device) comprising an electro-optical circuit package that includes a package substrate that includes at least one photonic integrated circuit (PIC) having at least one light source, a printed circuit board (PCB) including at least one optical connector to receive light from the at least one light source, and multiple liquid metal electrical contacts disposed between the package substrate and the PCB.

(54) In Example 2, the subject matter of Example 1 optionally includes liquid metal electrical contacts that are included on the package substrate and a PCB that includes multiple pins to contact the multiple liquid metal electrical contacts. The multiple pins include at least one power pin longer

than other pins of the multiple pins to provide power to the at least one light source.
(55) In Example 3, the subject matter of one or both of Examples 1 and 2 optionally includes a PCB socket to receive the electro-optical circuit package, wherein the liquid metal electrical contacts are included on the PCB socket and an electro-optical circuit package that includes multiple pins to contact the multiple liquid metal electrical contacts, including a power pin longer than other pins of the multiple pins to provide power to the at least one light source.
(56) In Example 4, the subject matter of one or any combination of Examples 1-4 optionally includes liquid metal electrical contacts that are included in an interposer contacting the package substrate.
(57) In Example 5, the subject matter of one or any combination of the Examples 1-4 optionally includes an integrated heat spreader contacting the PIC, a package socket on the PCB to receive the electro-optical circuit package, one or more removable tabs in contact with one or both of the package substrate and the integrated heat spreader, and one or more removable landing pads for the one or more removable frames on the package socket.
(58) In Example 6, the subject matter of Example 5 optionally includes a heat sink contacting the integrated heat spreader, and a clamp to attach the heat sink to the integrated heat spreader.
(59) In Example 7, the subject matter of one or any combination of Examples 1-6 optionally includes a waveguide connected to the at least one optical connector, and a light sensor coupled to the waveguide and configured to detect light emitted from the PIC.
(60) In Example 8, the subject matter of Example 7, including a positioning circuit operatively coupled to the waveguide and configured to produce one or more positioning control signals in response to light from the PIC.
(61) In Example 9, the subject matter of one or both of Examples 7 and 8 optionally includes a waveguide embedded in the PCB.
(62) In Example 10, the subject matter of one or any combination of Examples 1-9 optionally includes a light source of the PIC that is a vertical-emitting laser diode and the optical connector is arranged vertically from the electro-optical circuit package.
(63) In Example 11, the subject matter of one or any combination of Examples 1-9 optionally includes a light source of the PIC that is a side-emitting laser diode and the optical connector is arranged adjacent to the electro-optical circuit package.
(64) In Example 12, the subject matter of one or any combination of Examples 1-11 optionally includes a multi-zoned package socket including multiple movable socket zones, an electro-optical circuit package includes multiple light sources, and each movable socket zone including an optical connector aligned with a light source of the multiple light sources.
(65) In Example 13, the subject matter of one or any combination of Examples 1-11 optionally includes a multi-zoned package socket including multiple movable socket zones, at least one additional electro-optical circuit package that includes at least one light source, and each movable socket zone including at least one optical connector aligned with the at least one light source of the multiple light sources.
(66) Example 14 includes subject matter (such as a method of forming an electronic device) or can optionally be combined with one or any combination of Examples 1-13 to include such subject matter, comprising arranging a photonic integrated circuit (PIC) (that includes at least one light source) on a package substrate of an electro-optical circuit package, disposing multiple liquid metal electrical contacts on the package substrate, aligning position of the package substrate on a printed circuit board (PCB) by aligning light from the light source of the PIC and an optical connector included on the PCB, and attaching the aligned package substrate to the PCB.
(67) In Example 15, the subject matter of Example 14 optionally includes providing power to the light source of the PIC from the PCB when the PCB and the package substrate are separated by an alignment separation, and reducing the alignment separation to attach the package substrate to the PCB.

(68) In Example 16, the subject matter of Example 15 optionally includes sensing an optical signal from the light source of the PIC, producing one or more electrical control signals using the optical signal, and adjusting position of the package substrate relative to the PCB using a positioning mechanism responsive to the one or more electrical controls signals.
(69) In Example 17, the subject matter of one or any combination of Examples 14-16 optionally includes including an integrated heat spreader with the electro-optical circuit package, and attaching one or more removable tabs to at least one of the electro-optical circuit package or the integrated heat spreader.
(70) In Example 18, the subject matter of Example 17 optionally includes attaching the package substrate to a package socket of the PCB, and attaching one or more removable landing pads for the one or more removable tabs on the package socket.
(71) Example 19 include subject matter (such as an electronic system) or can optionally be combined with one or any combination of Examples 1-18 to include such subject matter, comprising at least one electro-optical circuit package including one or more photonic integrated circuits (PICs) that include one or more multiple light sources, a package substrate, multiple liquid metal electrical contacts on the package substrate; and a printed circuit board (PCB) including a package socket, wherein the package socket includes multiple socket zones, wherein each socket zone includes an optical connector aligned with a light source of the multiple light sources.
(72) In Example 20, the subject matter of Example 19 optionally includes multiple waveguides, including a waveguide connected to each of the optical connector; and multiple light sensors, including a light sensor coupled to each waveguide of the multiple waveguides, each light sensor configured to detect light emitted from a light source of the electro-optical circuit package.
(73) In Example 21, the subject matter of one or both of Examples 19 and 20 optionally include an electro-optical circuit package that includes an integrated heat spreader contacting the one or more PICs; a removable frame in contact with one or both of the package substrate and the integrated heat spreader; a first removable landing pad for the removable frame on the package socket; and a second removable landing pad for the removable frame on the PCB.
(74) In Example 22, the subject matter of one or any combination of Examples 19-21 optionally includes a PCB including multiple pins connected to the multiple liquid metal electrical contacts. The multiple pins including multiple power pins that provide power to the multiple light sources of the electro-optical circuit package, and the multiple power pins are longer than other pins of the PCB.
(75) In Example 23, the subject matter of one or any combination of Examples 19-22 optionally includes an array of multiple electro-optical circuit packages each electro-optical circuit package including at least one light source aligned with at least one optical connector of a socket zone.
(76) These non-limiting examples can be combined in any permutation or combination. The Abstract is provided to allow the reader to ascertain the nature and gist of the technical disclosure. It is submitted with the understanding that it will not be used to limit or interpret the scope or meaning of the claims. The following claims are hereby incorporated into the detailed description, with each claim standing on its own as a separate embodiment.

Claims

1. An electronic device comprising: an electro-optical circuit package including: at least one photonic integrated circuit (PIC) including at least one light source; and a package substrate; a printed circuit board (PCB) including at least one optical connector to receive light from the at least one light source; and multiple liquid metal electrical contacts disposed between the package substrate and the PCB.
2. The electronic device of claim 1, wherein the liquid metal electrical contacts are included on the package substrate; and wherein the PCB includes multiple pins to contact the multiple liquid metal

electrical contacts, including a power pin longer than other pins of the multiple pins to provide power to the at least one light source.
3. The electronic device of claim 1, including: a PCB socket to receive the electro-optical circuit package, wherein the liquid metal electrical contacts are included on the PCB socket; and wherein the electro-optical circuit package includes multiple pins to contact the multiple liquid metal electrical contacts, including a power pin longer than other pins of the multiple pins to provide power to the at least one light source.
4. The electronic device of claim 1, wherein the liquid metal electrical contacts are included in an interposer contacting the package substrate.
5. The electronic device of claim 1, including: an integrated heat spreader contacting the PIC; a package socket on the PCB to receive the electro-optical circuit package; one or more removable tabs in contact with one or both of the package substrate and the integrated heat spreader; and one or more removable landing pads for the one or more removable frames on the package socket.
6. The electronic device of claim 5, including: a heat sink contacting the integrated heat spreader; and a clamp to attach the heat sink to the integrated heat spreader.
7. The electronic device of claim 1, including: a waveguide connected to the at least one optical connector; and a light sensor coupled to the waveguide and configured to detect light emitted from the PIC.
8. The electronic device of claim 7, including a positioning circuit operatively coupled to the waveguide and configured to produce one or more positioning control signals in response to light from the PIC.
9. The electronic device of claim 7, wherein the waveguide is embedded in the PCB.
10. The electronic device of claim 1, wherein the light source of the PIC is a vertical-emitting laser diode and the optical connector is arranged vertically from the electro-optical circuit package.
11. The electronic device of claim 1, wherein the light source of the PIC is a side-emitting laser diode and the optical connector is arranged adjacent to the electro-optical circuit package.
12. The electronic device of claim 1, including: a multi-zoned package socket including multiple movable socket zones; wherein the electro-optical circuit package includes multiple light sources; and wherein each movable socket zone includes an optical connector aligned with a light source of the multiple light sources.
13. The electronic device of claim 1, including: a multi-zoned package socket including multiple movable socket zones; at least one additional electro-optical circuit package that includes at least one light source; and wherein each movable socket zone includes at least one optical connector aligned with the at least one light source of the multiple light sources.
14. A method of forming an electronic device, the method comprising: arranging a photonic integrated circuit (PIC) on a package substrate of an electro-optical circuit package, wherein the PIC includes at least one light source; disposing multiple liquid metal electrical contacts on the package substrate; aligning position of the package substrate on a printed circuit board (PCB) by aligning light from the light source of the PIC and an optical connector included on the PCB; and attaching the aligned package substrate to the PCB.
15. The method of claim 14, wherein the aligning the position of the package substrate includes: providing power to the light source of the PIC from the PCB when the PCB and the package substrate are separated by an alignment separation; and reducing the alignment separation to attach the package substrate to the PCB.
16. The method of claim 15, wherein the aligning the position of the package substrate includes: sensing an optical signal from the light source of the PIC; producing one or more electrical control signals using the optical signal; and adjusting position of the package substrate relative to the PCB using a positioning mechanism responsive to the one or more electrical controls signals.
17. The method of claim 14, including: including an integrated heat spreader with the electro-optical circuit package; and attaching one or more removable tabs to at least one of the electro-

optical circuit package or the integrated heat spreader.
18. The method of claim 17, including: wherein the attaching the package substrate to the PCB includes attaching the package substrate to a package socket of the PCB; and wherein the method further includes attaching one or more removable landing pads for the one or more removable tabs on the package socket.
19. An electronic system, the system comprising: at least one electro-optical circuit package including: one or more photonic integrated circuits (PICs) that include one or more multiple light sources; a package substrate; and multiple liquid metal electrical contacts on the package substrate; and a printed circuit board (PCB) including a package socket, wherein the package socket includes multiple socket zones, wherein each socket zone includes an optical connector aligned with a light source of the multiple light sources.
20. The system of claim 19, including: multiple waveguides, including a waveguide connected to each of the optical connector; and multiple light sensors, including a light sensor coupled to each waveguide of the multiple waveguides, each light sensor configured to detect light emitted from a light source of the electro-optical circuit package.
21. The system of claim 19, including: wherein the electro-optical circuit package includes an integrated heat spreader contacting the one or more PICs; a removable frame in contact with one or both of the package substrate and the integrated heat spreader; a first removable landing pad for the removable frame on the package socket; and a second removable landing pad for the removable frame on the PCB.
22. The system of claim 19, wherein the PCB includes multiple pins connected to the multiple liquid metal electrical contacts, including multiple power pins that provide power to the multiple light sources of the electro-optical circuit package, and wherein the power pins are longer than other pins of the PCB.
23. The system of claim 19, including an array of multiple electro-optical circuit packages each electro-optical circuit package including at least one light source aligned with at least one optical connector of a socket zone.